

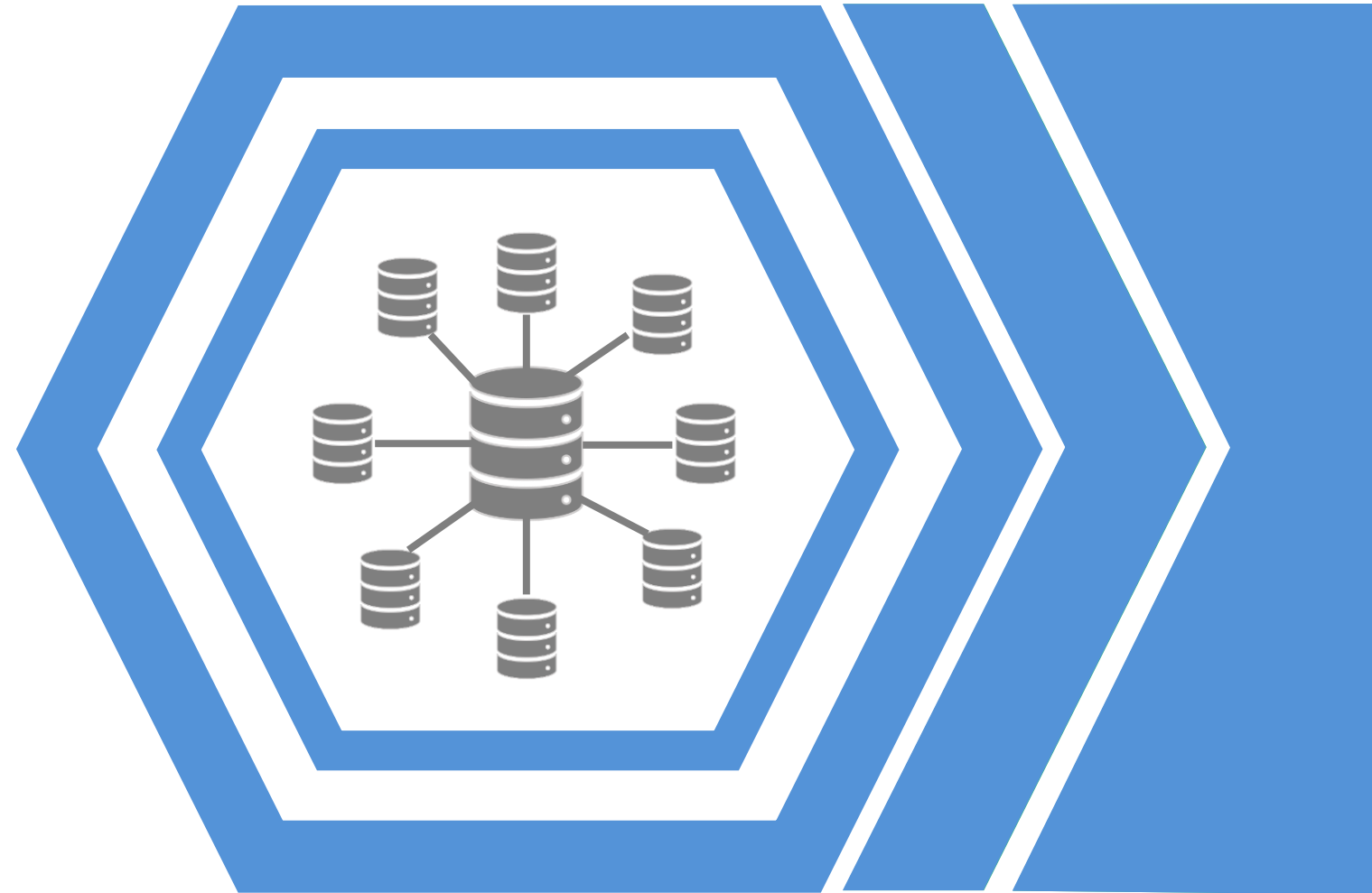


CSE 3020 Database System I

Lecture #37

Relational Calculus

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What will we learn today?

- Understanding Relational Model and Formal Query Language
- Relational Calculus
- Difference between relational algebra and relational calculus
- Tuple Calculus
- Quantifier
- Domain Calculus

Understanding Relational Model and Formal Query Language

Relational Model



RDBMS

Formal Query Language



SQL

- **Formal query languages** describing the relational model
 - Relational Algebra
 - Relational Calculus
- Forms the underlying structure for structured query language (SQL)

Relational Calculus

- one of formal query languages for the relational model
- no description of how to evaluate a query
- specifies **what is to be retrieved** rather than how to retrieve it
- **non-procedural language**
- Relational calculus can be expressed using
 - Tuple relational calculus
 - Domain relational calculus
- both forms specify a **retrieval request** using a single declarative expression **without specifying how and the order** of evaluation

Relational Calculus

- Many of the calculus expressions involves the use of **Quantifiers**.
- Quantifiers in the relational calculus is a determiner that determines the scope of the term it is attached to.
- There are two types of quantifiers:
 - **Universal Quantifiers**
 - **Existential Quantifiers**
- Relational calculus is important for two reasons:
 - Basis in mathematical logic
 - SQL has its foundations in the tuple relational calculus

Difference between relational algebra and relational calculus

Relational Algebra	Relational Calculus
Procedural , states how to obtain the result	Nonprocedural , states what results to be obtained
Order of operations is important	Order of operations is unimportant
Prescriptive , describes the steps to perform a task	Declarative , describes desired result
Queries are domain independent	Queries are domain dependent
Has operators to perform tasks	Has tuples and domain-oriented expressions

Tuple Calculus

- Based on specifying a number of tuple variables
- Each tuple variable ranges over a relation

$\{t \mid \text{COND}(t)\}$

where

t is a tuple variable

$\text{COND}(t)$ is a conditional BOOLEAN expression

- Result:
 - Set of all tuples that evaluate $\text{COND}(t)$ to TRUE

Tuple Calculus: Examples

EMPLOYEE (Fname, Minit, Lname, Ssn, Bdate, Address, Sex,
Salary, Super_ssn, Dno)

DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr_start_date)

DEPT_LOCATIONS (Dnumber, Dlocation)

PROJECT (Pname, Pnumber, Plocation, Dnum)

WORKS_ON (Essn, Pno, Hours)

DEPENDENT (Essn, Dependent_name, Sex, Bdate, Relationship)

Tuple Calculus: Examples

Example 1:

$\{t \mid$
 EMPLOYEE (t) **AND** t.Salary > 50000}

Example 2:

$\{t.Fname, t.Lname \mid$
 EMPLOYEE (t) **AND** t.Salary > 50000}

Example 3:

$\{t.Bdate, t.Address \mid$
 EMPLOYEE (t) **AND**
 t.Fname = 'John' **AND** t.Minit = 'B' **AND** t.Lname = 'Smith'}

Tuple Calculus – General Form

$$\{t_1.A_j, t_2.A_k, \dots, t_n.A_m \mid$$
$$\text{COND}(t_1, t_2, \dots, t_n, t_{n+1}, \dots, t_{n+m})\}$$

where

$t_1, t_2, \dots, t_n, t_{n+1}, \dots, t_{n+m}$ are **tuple variables**

each A_i is an attribute of the relation on which t_i ranges

COND is a condition or formula

Tuple Calculus – General Form: Example 1

For each employee, retrieve the employee's first and last name and
the first and last name of his or her immediate supervisor.

EMPLOYEE (Fname, Lname)

EMPLOYEE (Fname, Lname, Super_ssn)

{e.Fname, e.Lname, s.Fname, s.Lname |
EMPLOYEE(e) AND
EMPLOYEE(s) AND
e.Super_ssn=s.Ssn}

FName	LName	SSN	Super_SSN
Ann	Smith	12365 4789	88854 8623
Jeremy	Matthew	96968 7423	12365 4789
Peter	Williams	33326 5874	88854 8623
Elsa	David	88854 8623	null

Tuple Calculus: Examples

EMPLOYEE (Fname, Minit, Lname, Ssn, Bdate, Address, Sex,
Salary, Super_ssn, Dno)

DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr_start_date)

DEPT_LOCATIONS (Dnumber, Dlocation)

PROJECT (Pname, Pnumber, Plocation, Dnum)

WORKS_ON (Essn, Pno, Hours)

DEPENDENT (Essn, Dependent_name, Sex, Bdate, Relationship)

Tuple Calculus – Example 2

Retrieve the name and address of all employees who work for the

EMPLOYEE (Fname, Lname, Address) => Dno

‘Research’ department.

DEPARTMENT (Dname) => Dnumber

{e.Fname, e.Lname, e.Address |

EMPLOYEE(e) AND

(DEPARTMENT(d) AND

d.Dname = ‘Research’ AND

d.Dnumber = e.Dno) }

Quantifiers

➤ Two Types:

1. Universal Quantifiers
2. Existential Quantifiers

Quantifiers - Universal Quantifiers

➤ Universal Quantifiers:

- The universal quantifier denoted by $\forall$ (upside down A)
- It is read as for all which means that in a given set of tuples **exactly all tuples** satisfy a given condition
- is called the **universal or “for all” quantifier** because **every tuple** in “the universe of” tuples must make F true to make the quantified formula true.

Quantifiers - Existential Quantifiers:

➤ Existential Quantifiers:

- The existential quantifier denoted by $\exists$ (backward E)
- It is read as for all which means that in a given set of tuples there is **at least one occurrences** whose value satisfy a given condition
- It is called the **existential** or “**there exists**” quantifier because **any tuple** that exists in “the universe of” tuples may make F true to make the quantified formula true.

Quantifiers – Example 1

List the name of each employee who works on some project controlled by

EMPLOYEE (Fname, Lname) => Ssn WORKS_ON => Essn, Pno
department number 5.

PROJECT (Dnum) => Pnumber

EMPLOYEE (Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)

DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr_start_date)

DEPT_LOCATIONS (Dnumber, Dlocation)

PROJECT (Pname, Pnumber, Plocation, Dnum)

WORKS_ON (Essn, Pno, Hours)

DEPENDENT (Essn, Dependent_name, Sex, Bdate, Relationship)

Quantifiers – Example 1

List the name of each employee who works on *some* project controlled by

EMPLOYEE (Fname, Lname) => Ssn

WORKS_ON => Essn, Pno

department number 5.

PROJECT (Dnum) => Pnumber

{e.Lname, e.Fname |

EMPLOYEE(e) AND

(($\exists p$)($\exists w$) (PROJECT(p) AND

WORKS_ON(w) AND

p.Dnum = 5 AND

w.Essn = e.Ssn AND

p.Pnumber = w.Pno)))}

Quantifiers – Example 2

For every project located in ‘Stafford’, list the project number, the controlling department number, and the department manager’s last name, birth date, and address.

EMPLOYEE (Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)

DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr_start_date)

DEPT_LOCATIONS (Dnumber, Dlocation)

PROJECT (Pname, Pnumber, Plocation, Dnum)

WORKS_ON (Essn, Pno, Hours)

DEPENDENT (Essn, Dependent_name, Sex, Bdate, Relationship)

Quantifiers – Example 2

For every project located in ‘Stafford’, list the project number, the controlling department number, and the department manager’s last name, birth date, and address.

$\{p.Pnumber, p.Dnum, m.Lname, m.Bdate, m.Address \mid \text{PROJECT}(p)$
AND $\text{EMPLOYEE}(m)$ **AND** $p.Plocation = \text{‘Stafford’}$
AND $((\exists d)(\text{DEPARTMENT}(d)$
AND $p.Dnum = d.Dnumber$ **AND** $d.Mgr_ssn = m.Ssn))\}$

Quantifiers – Example 3

List the names of employees who have no dependents.

EMPLOYEE (Fname, Lname) => Ssn

DEPENDENT (Essn)

Existential Quantifiers:

{e.Fname, e.Lname |

EMPLOYEE(e) **AND**

(NOT (∃d) (DEPENDENT(d) **AND** e.Ssn = d.Essn)))}

Universal Quantifiers:

{e.Fname, e.Lname |

EMPLOYEE(e) **AND**

((∀d) (NOT(DEPENDENT(d)) **OR** NOT(e.Ssn=d.Essn))))}

Domain Calculus

- Domain calculus differs from tuple calculus in the type of variables used in formulas
- Rather than having variables range over tuples, the **variables range over single values from domains of attributes.**
- To form a relation of degree n for a query result, we must have n of these domain variables—one for each attribute.

Domain Calculus

- An expression of the domain calculus is of the form

$$\{x_1, x_2, \dots, x_n \mid \text{COND}(x_1, x_2, \dots, x_n, x_{n+1}, x_{n+2}, \dots, x_{n+m})\}$$

where

$x_1, x_2, \dots, x_n, x_{n+1}, x_{n+2}, \dots, x_{n+m}$ are **domain variables** that range over domains (of attributes)

COND is a condition or formula of the domain relational calculus.

Example for Domain Calculus

List the names of employees who have no dependents.

$\text{EMPLOYEE (Fname, Lname) } \Rightarrow \text{ Ssn}$

DEPENDENT (Essn)

Tuple Calculus:

$\{e.\text{Fname}, e.\text{Lname} \mid \text{EMPLOYEE}(e) \text{ AND}$

$(\text{NOT } (\exists d)(\text{DEPENDENT}(d) \text{ AND } e.\text{Ssn} = d.\text{Essn}))\}$

Domain Calculus:

$\{F, L \mid (\exists S)(\text{EMPLOYEE}(F, M, L, S, B, A, SE, SA, SU, D) \text{ AND}$

$(\text{NOT } (\exists E)(\text{DEPENDENT}(E,D,N,S,B,R) \text{ AND } S=E))\}$

Example1 for Tuple Calculus

Student (ID, Name, Major)

Course (Dept, Num, Title)

Dept (Abbrev, Name, Office)

Enroll (SID, Dept, Num, Date)

Express the following English queries in Tuple Relational Calculus:

1. What are the department abbreviations and titles of all courses numbered 101?
2. What are the IDs of the MATH students taking “Discrete Structures?”

Example1 for Tuple Calculus – Answers

Student (ID, Name, Major)

Course (Dept, Num, Title)

Dept (Abbrev, Name, Office)

Enroll (SID, Dept, Num, Date)

1. What are the department and titles of all courses numbered 101?

Course (Dept, Title)

Course (Num)

{c.Dept, c.Title |

Course(c) AND

c.Num = 101}

Example1 for Tuple Calculus – Answers

Student (ID, Name, Major)

Course (Dept, Num, Title)

Dept (Abbrev, Name, Office)

Enroll (SID, Dept, Num, Date)

2. What are the IDs of the MATH students taking “Discrete Structures?”

Student (ID)

Student (Major)

Course (Title) => Dept, Num

Enroll (SID, Dept, Num)

Example1 for Tuple Calculus – Answers

2. What are the IDs of the MATH students taking “Discrete Structures?”

Student (ID)

Student (Major)

Course (Title) => Dept, Num

Enroll (SID, Dept, Num)

{s.ID |

Student(s) **AND** s.Major = ‘MATH’ **AND**

($\exists e$)(Enroll(e) **AND** s.ID = e.SID **AND**

($\exists c$)(Course(c) **AND**

e.Dept = c.Dept **AND** e.Num = c.Num **AND**

c.Title = ‘Discrete Structures’))}

Example2 for Tuple Calculus

For each of these TRC queries, what is the corresponding query in conversational English?

1. $\{s.Name, s.ID \mid \text{Student}(s) \text{ AND } s.Major = \text{'CSE'}\}$
2. $\{e.Date \mid \text{Enroll}(e) \text{ AND } (\exists c)(\text{Course}(c) \text{ AND } e.Dept = c.Dept \text{ AND } e.Num = c.Num \text{ AND } c.Title = \text{'Database Design'})\}$

Example2 for Tuple Calculus – Answers

1. $\{s.Name, s.ID \mid$

Student(s) **AND**

$s.Major = 'CSE'\}$

Answer:

What are the names and IDs of students from the CSE majors?

Example2 for Tuple Calculus – Answers

2. $\{e.Date \mid$
 $Enroll(e) \text{ AND}$
 $(\exists c) (Course(c) \text{ AND}$
 $e.Dept = c.Dept \text{ AND } e.Num = c.Num \text{ AND}$
 $c.Title = 'Database Design')\}$

Answer:

On which dates did students enroll in “Database Design?”